

Name:

Total Points: 15

Roll number:

Duration: 60 minutes

1. [CO1, CO2, CO4] k-NN Classification (0.5 + 0.5 points)

A k-NN classifier with $k = 7$ is trained on a binary classification dataset where 90% of samples belong to class A and 10% to class B . For a query point Q , the 3 nearest neighbours belong to class B and the 4 next-nearest neighbours belong to class A , with Q being geometrically closer to its class- B neighbours. The classifier predicts class A for Q . What is the reason, and what limitation of k-NN does this highlight?

- (a) k-NN counts votes equally among all k neighbours, regardless of distance. Four class- A votes outnumber three class- B votes, a problem made worse by the imbalanced training distribution.
- (b) k-NN assigns higher weight to closer neighbours by default, so the class- A prediction suggests the training data is noisy rather than a flaw in the algorithm.
- (c) A 4 vs. 3 vote split is treated as a tie in k-NN, and the outcome depends on the tie-breaking rule used in the implementation.
- (d) k-NN accounts for class frequency during prediction, so the 90% share of class A overrides the local evidence around Q .

Solution:

Option (A) is correct. Standard k-NN assigns one vote to each neighbour regardless of distance. Here, 4 class- A votes outnumber 3 class- B votes, so Q is assigned to class A even though it lies closer to class- B points. This highlights a fundamental limitation: in an imbalanced dataset, majority-class points appear so frequently in feature space that they routinely enter the neighbourhood of minority-class query points, consistently outvoting the minority class irrespective of local geometry.

2. [CO1, CO2, CO4] Decision Trees (0.5 + 0.5 points) In a decision tree, each node split is defined by a linear combination of features rather than a single feature threshold. How does this change the geometry of the decision boundary?

- (a) The decision boundary remains piecewise axis-aligned, but each piece now spans a higher-dimensional subspace.
- (b) The tree partitions feature space using oblique/tilted hyperplanes, allowing boundaries that are not restricted to splits along a single feature.
- (c) The hypothesis class of the tree becomes equivalent to that of a linear classifier trained on the same features.
- (d) The tree can represent any convex decision boundary exactly, since each linear split divides feature space into two regions.

Solution:

Option (B) is correct. A standard decision tree splits on one feature at a time, so its boundaries are always parallel to the coordinate axes. Using $\mathbf{w}^\top \mathbf{x} \leq t$ allows each split to combine multiple features, so the boundary can be tilted at any angle. A single-feature split is just the special case where only one entry of $\mathbf{w}$ is non-zero.

3. [CO1, CO2, CO4] Ensemble Learning (3 points)

A hospital deploys a single decision tree to classify patient scans as malignant or benign. The model performs well on training data but makes inconsistent errors on new patients sometimes misclassifying clear cases. A machine learning engineer proposes replacing it with an ensemble of multiple classifiers trained on different subsets of patient data.

- (a) Explain why the ensemble of classifiers is likely to perform better than the single decision tree on new patient data in this scenario.
- (b) A single classifier can fail for three fundamental reasons. Name each failure type, describe the condition under which it arises, and state the consequence it has on the learning algorithm. For each failure, also briefly explain how ensemble learning addresses it.

Solution:**Part (a):**

The single decision tree makes inconsistent errors on new patients, which indicates high variance: the model is sensitive to the specific training samples it has seen and does not generalise reliably. When multiple classifiers are trained on different subsets of patient data, each model learns slightly different patterns and makes different errors on new inputs. This is referred to as diversity among classifiers, and it is the key condition for ensembling to be effective. When the predictions of these diverse classifiers are combined for example, through majority voting the errors made by individual classifiers tend to cancel out, since they do not all fail on the same input at the same time. The correct prediction, being more consistent across classifiers, accumulates more votes and becomes the final output. As a result, the combined model makes fewer errors on new data than any single classifier would. In the hospital setting, this means fewer misclassifications of patient scans, which directly improves diagnostic reliability.

Part (b):

1. Statistical Failure: This arises when the training dataset is too small relative to the hypothesis space. Several different hypotheses perform similarly on training data, making it difficult to identify the correct one. The consequence is high variance the chosen hypothesis may not generalise to new data. Ensemble learning addresses this by averaging the votes of multiple hypotheses, reducing the risk of committing to a single wrong one.

2. Computational Failure: This arises when the learning algorithm relies on local search and gets stuck in a local minimum, failing to find the best hypothesis even when sufficient data is available. The consequence is high computational variance. Ensemble learning addresses this by running multiple learners from different starting points; their combination is more likely to approximate the true hypothesis than any single run.

3. Representational Failure: This arises when the true relationship between the input and output cannot be expressed by any hypothesis in the chosen model family. No amount of data or computation can resolve this. The consequence is high bias. Ensemble learning addresses this by combining multiple model families or hypothesis classes, effectively expanding the space of representable functions beyond what any single model family could capture.

4. [CO1, CO2, CO4] Kernel SVC**(3 points)**

- (a) What are the necessary conditions for a valid kernel function?
- (b) Given the kernel functions, define the transformation function for each of the following:
 - i. $K(\bar{u}, \bar{v}) = \cos u_1 \cos v_1 + \sin u_1 \sin v_1$
 - ii. $K(\bar{u}, \bar{v}) = (u_1 v_1 + u_2 v_2)^2$
 - iii. $K(\bar{u}, \bar{v}) = 1 + u_1 v_1 + u_2 v_2 + u_1 u_2 v_1 v_2$
 - iv. $K(\bar{u}, \bar{v}) = (u_1 v_1 + u_2 v_2)^2 + 2(u_1 v_1 + u_2 v_2) + 1$

Solution:

- (a) For a function K to be a valid kernel, the two necessary and sufficient conditions are:
 - i. **Symmetry:** Symmetric to its arguments

$$K(x_1, x_2) = K(x_2, x_1)$$

- ii. **Positive (Semi) Definiteness:**

$$\mathbf{z}^T K \mathbf{z} \geq 0 \quad \forall \text{ non-zero } \mathbf{z}$$

Such a kernel is said to satisfy **Mercer's condition**.

- (b) (i)

$$\phi(\bar{u}) = \begin{bmatrix} \cos u_1 \\ \sin u_1 \end{bmatrix}$$

- (ii) Expanding: $u_1^2 v_1^2 + u_2^2 v_2^2 + 2u_1 u_2 v_1 v_2$

$$\phi(\bar{u}) = \begin{bmatrix} u_1^2 \\ u_2^2 \\ \sqrt{2} u_1 u_2 \end{bmatrix}$$

(iii) By inspection:

$$\phi(\bar{u}) = \begin{bmatrix} 1 \\ u_1 \\ u_2 \\ u_1 u_2 \end{bmatrix}$$

(iv) Recognising the perfect square: $K = (1 + u_1 v_1 + u_2 v_2)^2$. Expanding:

$$K = 1 + u_1^2 v_1^2 + u_2^2 v_2^2 + 2u_1 v_1 + 2u_2 v_2 + 2u_1 u_2 v_1 v_2$$

$$\phi(\bar{u}) = \begin{bmatrix} 1 \\ u_1^2 \\ u_2^2 \\ \sqrt{2} u_1 \\ \sqrt{2} u_2 \\ \sqrt{2} u_1 u_2 \end{bmatrix}$$

5. [CO1, CO2, CO4] Naive Bayes Classifier

(3 points)

A dataset of 14 instances is collected to predict whether a person buys a computer (Yes/No) based on two categorical features Age $\in \{\text{Young, Middle, Senior}\}$ and Income $\in \{\text{Low, Medium, High}\}$:

Age	Income	Buys Computer
Young	High	No
Young	High	No
Middle	High	Yes
Senior	Medium	Yes
Senior	Low	Yes
Senior	Low	No
Middle	Low	Yes
Young	Medium	No
Young	Low	Yes
Senior	Medium	Yes
Young	Medium	Yes
Middle	Medium	Yes
Middle	High	Yes
Senior	Medium	No

(a) State the Naive Bayes decision rule for this problem and identify which conditional independence assumption it makes. Why might this assumption be violated for Age and Income in practice?

(b) Predict whether a Middle-aged person with a High income will buy a computer. Show all prior and likelihood calculations.

Solution:

(a) The Naive Bayes decision rule is:

$$\hat{y} = \arg \max_{c \in \{\text{Yes, No}\}} p(c) \prod_i p(x_i | c)$$

The assumption is **conditional independence** of features given the class: $p(\text{Age, Income} | c) = p(\text{Age} | c) \cdot p(\text{Income} | c)$. In practice this is violated because Age and Income are correlated e.g., Senior individuals are more likely to have Medium/High income.

(b) Class counts: Yes = 9, No = 5, Total = 14.

$$p(\text{Yes}) = \frac{9}{14}, \quad p(\text{No}) = \frac{5}{14}$$

$$p(\text{Middle} | \text{Yes}) = \frac{4}{9}, \quad p(\text{Middle} | \text{No}) = \frac{0}{5}$$

$$p(\text{High} | \text{Yes}) = \frac{2}{9}, \quad p(\text{High} | \text{No}) = \frac{2}{5}$$

Unnormalized posteriors:

$$p(\text{Yes}) \cdot p(\text{Middle}|\text{Yes}) \cdot p(\text{High}|\text{Yes}) = \frac{9}{14} \times \frac{4}{9} \times \frac{2}{9} = \frac{72}{1134} \approx 0.0635$$
$$p(\text{No}) \cdot p(\text{Middle}|\text{No}) \cdot p(\text{High}|\text{No}) = \frac{5}{14} \times \frac{0}{5} \times \frac{2}{5} = 0$$

MAP prediction: Yes (0.0635 > 0)

This course is offered in two tracks: **3XX (undergraduate)** and **5XX (graduate)**. Attempt **only one** of the following questions based on the course code written in the question.

Note: Attempting both questions or the question not corresponding to your course code **will not** fetch extra marks.

6. **[CO1, CO2, CO4] Decision Trees [3XX]** (4 points)

The following dataset records whether a patient is admitted to the ICU based on two features Age $\in \{\text{Young, Old}\}$ and BP (Blood Pressure) $\in \{\text{Low, High}\}$ along with a continuous outcome stay (days in hospital):

Instance	Age	BP	Stay (days)	Admitted
1	Young	Low	3	No
2	Young	Low	5	No
3	Young	High	8	Yes
4	Old	Low	6	Yes
5	Old	High	12	Yes
6	Old	High	10	Yes
7	Young	High	7	Yes
8	Old	Low	4	No

Using the admitted column as the target.

(a) Compute $Gini_{split}$ for both Age and BP and identify the root splitting attribute.

(b) Construct the decision tree up to the 2nd split by applying the Gini criterion recursively. Show the Gini calculation at each internal node.

(c) A new patient is Young with High BP. Trace their path through the tree built in (b) and state the predicted class. If instance 3 had Admitted = No, would the root split change? Justify without recomputing.

Solution:

(a) Class counts: Yes = {3,4,5,6,7}, No = {1,2,8}. Total $|S| = 8$.

Split by Age:

$$S_{\text{Young}} = \{1, 2, 3, 7\} : \quad 2 \text{ Yes}, 2 \text{ No} \quad \Rightarrow \text{Gini} = 1 - \left(\frac{2}{4}\right)^2 - \left(\frac{2}{4}\right)^2 = 0.5$$

$$S_{\text{Old}} = \{4, 5, 6, 8\} : \quad 3 \text{ Yes}, 1 \text{ No} \quad \Rightarrow \text{Gini} = 1 - \left(\frac{3}{4}\right)^2 - \left(\frac{1}{4}\right)^2 = 0.375$$

$$Gini_{split}(\text{Age}) = \frac{4}{8}(0.5) + \frac{4}{8}(0.375) = 0.4375$$

Split by BP:

$$S_{\text{Low}} = \{1, 2, 4, 8\} : \quad 1 \text{ Yes}, 3 \text{ No} \quad \Rightarrow \text{Gini} = 1 - \left(\frac{1}{4}\right)^2 - \left(\frac{3}{4}\right)^2 = 0.375$$

$$S_{\text{High}} = \{3, 5, 6, 7\} : \quad 4 \text{ Yes}, 0 \text{ No} \quad \Rightarrow \text{Gini} = 0$$

$$Gini_{split}(\text{BP}) = \frac{4}{8}(0.375) + \frac{4}{8}(0) = 0.1875$$

Root attribute: BP (0.1875 < 0.4375)

(b) *1st split* — BP:

- BP = High $\Rightarrow S = \{3, 5, 6, 7\}$: pure (all Yes) $\Rightarrow$ **leaf: Yes**
- BP = Low $\Rightarrow S = \{1, 2, 4, 8\}$: 1 Yes, 3 No $\Rightarrow$ impure, apply 2nd split

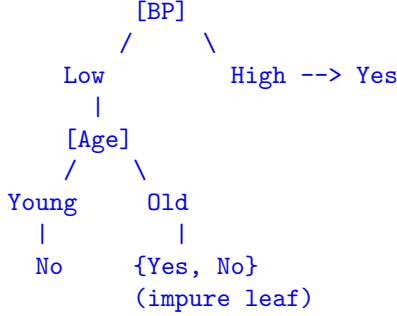
2nd split on $S_{\text{Low}} = \{1, 2, 4, 8\}$, only remaining attribute is Age:

$$S_{\text{Young}} = \{1, 2\} : \quad 0 \text{ Yes, } 2 \text{ No} \quad \Rightarrow \text{Gini} = 0$$

$$S_{\text{Old}} = \{4, 8\} : \quad 1 \text{ Yes, } 1 \text{ No} \quad \Rightarrow \text{Gini} = 0.5$$

$$\text{Gini}_{\text{split}}(\text{Age} \mid \text{BP}=\text{Low}) = \frac{2}{4}(0) + \frac{2}{4}(0.5) = 0.25$$

Tree after 2nd split:



(c) Young, High BP $\Rightarrow$ BP = High $\Rightarrow$ **Predicted: Yes** (pure leaf at 1st split).

6. [CO1, CO2, CO4] **Decision Trees [5XX]**

(4 points)

Use the Stay column as the target. The splitting criterion is weighted variance reduction:

$$\Delta \text{Var}(S, A) = \text{Var}(S) - \sum_v \frac{|S_v|}{|S|} \text{Var}(S_v), \quad \text{where} \quad \text{Var}(S) = \frac{1}{|S|} \sum_i (y_i - \bar{y})^2$$

(a) Compute ΔVar for both Age and BP and identify the root splitting attribute.

(b) Apply one further split on the larger variance partition from (a). State the predicted value (leaf mean) at each resulting leaf.

(c) A new patient is Old with Low BP. Trace their path through the tree built in (b) and state the predicted stay. If instance 4 had Stay = 11, would the root split change? Justify without recomputing.

Solution: (a) Stay values: $S = \{3, 5, 8, 6, 12, 10, 7, 4\}$, $\bar{y} = 55/8 = 6.875$

$$\text{Var}(S) = \frac{1}{8} \sum (y_i - 6.875)^2 = 8.113$$

Split by Age:

$$S_{\text{Young}} = \{3, 5, 8, 7\}, \quad \bar{y} = 5.75, \quad \text{Var} = 3.6875$$

$$S_{\text{Old}} = \{6, 12, 10, 4\}, \quad \bar{y} = 8, \quad \text{Var} = 10$$

$$\Delta \text{Var}(\text{Age}) = 8.113 - \left[\frac{4}{8}(3.6875) + \frac{4}{8}(10) \right] = 1.269$$

Split by BP:

$$S_{\text{Low}} = \{3, 5, 6, 4\}, \quad \bar{y} = 4.5, \quad \text{Var} = 1.25$$

$$S_{\text{High}} = \{8, 12, 10, 7\}, \quad \bar{y} = 9.25, \quad \text{Var} = 3.6875$$

$$\Delta \text{Var}(\text{BP}) = 8.113 - \left[\frac{4}{8}(1.25) + \frac{4}{8}(3.6875) \right] = 5.644$$

Root attribute: BP (5.644 > 1.269)

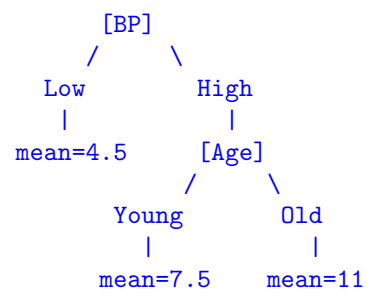
(b) The larger variance partition after the BP split is $S_{\text{High}} = \{8, 12, 10, 7\}$ ($\text{Var} = 3.6875 > 1.25$). Split by Age:

$$S_{\text{Young, High}} = \{8, 7\} \Rightarrow \text{leaf, mean} = 7.5$$

$$S_{\text{Old, High}} = \{12, 10\} \Rightarrow \text{leaf, mean} = 11$$

$S_{\text{Low}} = \{3, 5, 6, 4\}$ is not further split $\Rightarrow$ **leaf, mean** = 4.5

Tree after 2nd split:



(c) Old, Low BP $\Rightarrow$ BP = Low $\Rightarrow$ **leaf, predicted stay = 4.5 days** (2nd split not reached).
